I/O MODULE ORGANIZATION

PREPARED BY: ENGR. GLA

INPUT/OUTPUT ORGANIZATION

- THE I/O ORGANIZATION OF THE COMPUTER, PROVIDES AN EFFICIENT MODE OF COMMUNICATION BETWEEN THE **CENTRAL SYSTEM** AND THE **OUTSIDE ENVIRONMENT**.
- THE INPUT/OUTPUT ORGANIZATION OF COMPUTER DEPENDS UPON THE **SIZE** OF COMPUTER AND THE **PERIPHERALS** CONNECTED TO IT.

PERIPHERAL DEVICES

- AN **EXTERNAL DEVICE** CONNECTED TO AN I/O MODULE WHICH PROVIDE A MEANS OF **EXCHANGING DATA** BETWEEN THE **EXTERNAL DEVICE** ENVIRONMENT AND THE **COMPUTER**
- THERE IS A **LINK** USED TO **EXCHANGE CONTROL, STATUS, AND DATA** BETWEEN THE **I/O MODULE** AND THE **EXTERNAL DEVICE.**

INPUT/OUTPUT INTERFACE

- INPUT/OUTPUT INTERFACE PROVIDES A METHOD FOR **TRANSFERRING INFORMATION** BETWEEN **INTERNAL STORAGE** AND **EXTERNAL I/O DEVICES.**
- PERIPHERALS CONNECTED TO A COMPUTER NEED **SPECIAL COMMUNICATION LINKS** FOR **INTERFACING** THEM WITH THE **CENTRAL PROCESSING UNIT**.
- THE **PURPOSE** OF **COMMUNICATION LINK** IS TO **RESOLVE THE DIFFERENCES** THAT EXIST BETWEEN THE **CPU** AND EACH **PERIPHERALS**.

MAJOR DIFFERENCES OF CPU AND PERIPHERALS

- THE **DATA TRANSFER RATE** OF PERIPHERALS IS USUALLY **SLOWER** THAN THE TRASNFER RATE OF CPU AND CONSEQUENTLY, A **SYNCHRONOZATION MECHANISM** MAY BE NEEDED.
- **DATA CODES** AND **FORMATS** IN THE PHERIPERALS DIFFER FORM THE WORD FORMAT IN THE CPU AND MEMORY

  PERIPHERALS – BLOCK

  CPU OR MEMORY – BITS, BYTE, WORD

MAJOR DIFFERENCES OF CPU AND PERIPHERALS

- THE **OPERATING MODES** OF PHERIPHERALS ARE DIFFERENT FROM EACH OTHER AND MUST BE CONTROLLED SO AS NOT TO DISTURB THE OPERATION BETWEEN OTHER PERIPHERALS AND THE CPU.

INPUT/OUTPUT INTERFACE

- TO RESOLVE THESE DIFFERENCES, COMPUTER SYSTEMS INCLUDE SPECIAL HARDWARE COMPONENTS BETWEEN THE CPU AND PERIPHERALS TO SUPERVISE AND SYNCHRONIZE ALL INPUT AND OUTPUT TRANSFERS. THESE COMPONENTS ARE CALLED **INTERFACE UNITS.**

PERIPHERAL DEVICES

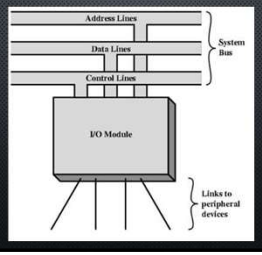
TWO CATEGORIES OF EXTERNAL DEVICES:

1. **HUMAN READABLE –** SUITABLE FOR COMMUNICATING WITH THE COMPUTER USER.

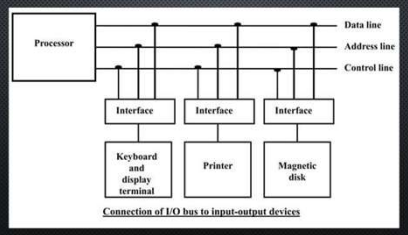
   EXAMPLES: PRINTERS, KEYBOARD, MONITOR, MOUSE
2. **MACHINE READABLE** – SUITABLE FOR COMMUNICATING WITH EQUIPMENT

   EXAMPLES: SENSORS AND ACTUATORS

GENERIC MODEL OF I/O MODULE



I/O BUS MODULE AND INTERFACE MODULE



Connection of I/O bus to input-output devices

MODULE FUNCTION

THE MAJOR FUNCTIONS OR REQUIREMENTS FOR AN I/O MODULE FALL INTO THE FOLLOWING:

- CONTROL AND TIMING
- PROCESSOR COMMUNICATION
- DEVICE COMMUNICATION
- DATA BUFFERING

CONTROL AND TIMING

1. THE PROCESSOR INTERROGATES THE I/O MODULE TO CHECK THE STATUS OF THE ATTACHED DEVICE.
2. THE I/O MODULE RETURNS THE DEVICE STATUS
3. IF THE DEVICE IS OPERATIONAL AND READY TO TRANSMIT, THE PROCESSOR REQUESTS THE TRASNFER OF DATA, BY MEANS OF A COMMAND TO THE I/O MODULE
4. THE I/O MODULE OBTAINS A UNIT OF DATA (E.G 8 OR 16 BITS) FROM THE EXTERNAL DEVICES
5. THE DATA ARE TRANSFERRED FROM THE I/O MODULE TO THE PROCESSOR

PROCESSOR COMMUNICATION

COMMAND DECODING:

THE I/O MODULE ACCEPTS COMMANDS FROM THE PROCESSOR, TYPICALLY SENT AS SIGNALS ON THE CONTROL BUS.

DATA:

DATA ARE EXCHANGED BETWEEN THE PROCESSOR AND THE I/O MODULE OVER THE DATA BUS.

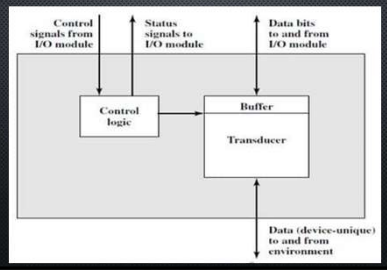
STATUS REPORTING:

PERIPHERALS ARE OFTEN SLOW, IT'S IMPORTANT TO KNOW THE STATUS THROUGH THE I/O MODULE. GENERATES STATUS SIGNAL (BUSY, READY, IDLE).

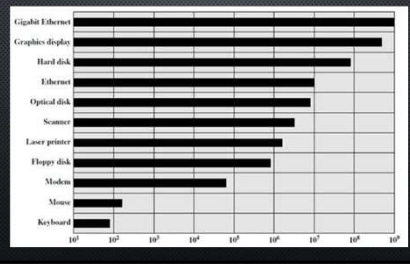
ADDRESS RECOGNITION:

JUST AS EACH WORD OF MEMORY HAS AN ADDRESS, SO DOES EACH I/O DEVICE.

DEVICE COMMUNICATION



DATA BUFFERING



ACTIVITY 3

1. IN YOUR OWN UNDERSTANDING, WHAT CAUSES INCOMPATIBILITY BETWEEN THE COMPUTER SYSTEM AND PERIPHERALS?
2. EXPLAIN THE FUNCTION OF THE INTERFACE UNITS.
3. EXPLAIN THE STRUCTURE OF A DEVICE COMMUNICATION